Week 6 Progress

Explainable Multi-Label Content-Based Image Retrieval for Aerial Images using Classical Computer Vision

Group 16

Dev Patel – AU2440252
Aashi Bhimjiyani – AU2440027
Aastha Gandhi – AU2440107
Dhruvi Faldu – AU2440216

1. Aim of Week 6

- Do the specific contexts (without labels) in the neighbourhoods lower the abottom of V3 (frozen) in the frozen state.
- There were two approaches that were attempted: Average Query Expansion (AQE) and k-reciprocal re-ranking.
- There were no changes to the features and weights of the V3.

2. Frozen V3 Baseline

- The descriptors used are: HSV, LBP (radius 1), LBP (radius 2) and HOG.
- Fusion weights: HSV 0.20, LBP-r1 0.40, LBP-r2 0.20, HOG 0.20.
- Accurately replicated V3 results before starting

3. Evaluation Setup

- The same 5 fold split of the training images as previously used.
- The number of queries that have been made: 480, # of images in the gallery: 1920
- It wasn't the 600 test images that were not used.
- No labels were attached to the pictures to be retrieved, re-ranked or selected parameters.
- Only labels were used after for calculating metrics.

4. Average Query Expansion (AQE)

- An average of top results of a query was computed to create a new query.
- Selection Score changed by only +0.0023.
- Class P@10 played 5 games, and won 4 and lost 5 games.
- Only preserved as an ablation.

5. Gallery-Only k-reciprocal Re-ranking

- Only images of the reciprocal neighbours in the gallery were used in the construction of the reciprocal neighbours.
- Parameters tested: k1 (10 to 40), k2 (3, 5, 8) and lambda (0.2 to 0.9).
- The same setting was chosen in all 5 folds: k1 = 30, k2 = 3, lambda = 0.5.
- Note that for comparison only, a version with the other queries (transductive) was run.
- The transductive version was not used as it is not possible for a real user to send more than one query at a time.

6. Week 6 Results

System	Class P@10	Class mAP	ML mAP	Selection score
V3 baseline	0.5069	0.2547	0.2738	0.4439
AQE (ablation)	0.5028	0.2618	0.2774	0.4462
Transductive KR (ablation)	0.5474	0.3231	0.2938	0.4698
Gallery-only KR (V4)	0.5361	0.3206	0.2938	0.4686

- The Selection Score is improved by +0.0247 with Gallery-only k-reciprocal.
- Class mAP went from 0.2547 to 0.3206.
- Class P@10 went from 0.5069 to 0.5361.
- ML mAP went from 0.2738 to 0.2938.
- ML P@10 and nDCG@10 changed only a little (+0.0070 and +0.0012).
- In all five folds there was positive progress.
- The improvements were the greatest in playground and bridge and square is still poor.

7. Decision

- Rule 1: If Selection Score is increased by more than +0.005 then the Rule 1 applies. Passed (+0.0247).
- Rule 2: At least a +4 in 5 folds. Passed (5 of 5).
- The Rule 3 states that the class P@10 should not be downgraded. Passed (+0.0292).
- V4 = V3 + k-reciprocal re-ranking method was used.

8. Challenges Faced

- There was a slight increase in AQE.
- It's not a top 10 multiple ranking label gain, it's an increase in a class level, it's a full ranking mAP observed.
- The relationships of gallery must be done beforehand before re-ranking.
- The transductive version is a little bit better, but with other queries which we were not able to provide.

9. Week 6 Conclusion

In the following, outside of Week 6, we studied if neighborhood information can be used to complement the base V3 unlabelled. This is not the case with AQE, but rather, gallery-only k-reciprocal re-ranking was found to be a clear and consistent improvement in a number of classes, particularly Class mAP and Class P@10, and was chosen as V4. The improvement is not as pronounced for the top-10 multi-label ranking and it is still possible to maintain the improvement for the worst classes. This week no test set was used.

Week 7 Progress Plan

Objective

The primary aim for week 7 is to test on test set V4; consider all the weak classes.

1. Final Evaluation

- V4_600 is a series of test pictures that will be used to run V4 once.
- Be sure to compare to the best model that has been done the year before.

2. Analysis

- Re-check results per class particularly square, bridge and airport.

3. Integration

- Implement re-ranking in demo and maintain explanation & reliability output.

Expected Outcome of Week 7

V4 should be complete by the end of week 7 and completed with a final test result, and a new demo.

Note: Only at the end of the test, 600 test images were used, for the final assessment.